MIL-F-11673F(EA)
17 December 1987
SUPERSEDING
MIL-F-11673E(MU)
21 October 1970

MILITARY SPECIFICATION

FUZE, SMOKE POT, FLOATING, ELECTRIC, M209

This specification is approved for use within U.S. Army Chemical Research, Development and Engineering Center Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of fuze.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

TT-W-572 - Wood-Preservative, Mater-Repellent.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-F-320 - Fiberboard: Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, U.S. Army Chemical Research, Development and Engineering Center, ATTN: SMCCR-SPT-S, Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter. :

AMSC N/A FSC 1365

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MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-B-2427 - Boxes, Ammunition Packing; Wood, Nailed.
MIL-B-46506 - Boxes, Ammunition Packing, Wood, Wirebound.

STANDARDS

FEDERAL

FED-STD-123 - Marking for Shipment (Civil Agencies).

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-1168 - Ammunition Lot Numbering.
MIL-STD-1234 - Pyrotechnics: Sampling, Inspection and Testing.

2.1.2 Other Government documents and drawings. The following other Government documents and drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

CODE OF FEDERAL REGULATIONS

49 CFR 171 to 179 - Department of Transportation Hazardous Materials Regulations.

(The Code of Federal Regulations is avaiable from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. Orders for the above Publication should cite “49 CFR 171 to 179.”)

DRAWINGS

U.S. ARMY CHEMICAL RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

36-7-25 - Fuze, Smoke Pot, Floating, Electric, M209.

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

19-48-4116/100AR1 - Unitization Procedures for Boxed Ammunition and Components on 4-Way Entry Pallets.

(Copies of specifications, standards, drawings, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

Uniform Freight Classification Rules

(Application for copies should be addressed to Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.)

National Motor Freight Classification Rules

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

2.3 Order of Precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing 36-7-25 or on the subsidiary drawings shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the fuze shall conform to the specifications and drawings listed on Drawing 36-7-25 and on subsidiary drawings.

3.2 Assembly The fuze shall be assembled as shown on Drawing 36-7-25.

3.3 Functioning. The fuze shall exhibit a minimum flash length of 6 inches when tested in accordance with 4.2.4.1.

3.4 Electrical resistance. The electrical resistance of the fuze shall be as specified in table I when tested in accordance with 4.2.4.2.

3.5 Dryness. The moisture and volatile content of ignition mix (Drawing 143-8-2) shall not exceed 0.5 percent when tested in accordance with 4.2.4.3.

3.6 Workmanship. The fuze shall be free from damage such as dents, leaks, cracks and deformation which would impair its intended use.

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TABLE I. Electrical resistance

Employed squib	Resistance of fuze
S-67	0.85 ± 0.25 ohm
S-65	1.55 ± 0.50 ohms

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements,

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 for which specific inspection has not been provided in this specification have been satsified.

4.2 Quality conformance inspection.

4.2.1 Lotting. Each lot shall be identified and controlled in accordance with MIL-STD-1168. Contained in any one interfix number of fuzes, no more than one interfix number from one manufacturer of the following components shall be permitted.

a. Ignition mix
b. Electric squibs

4.2.2 Sampling.

4.2.2.1 For examination. Sampling shall be conducted in accordance with MIL-STD-105. Sampling for in-process inspection of ignition mix shall be conducted in accordance with MIL-STD-1234.

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4.2.2.2 For destructive test. Sampling shall be conducted in accordance with MIL-STD-105 using level S-4.

4.2.3 Inspection procedure.

4.2.3.1 For examination. Sample fuzes and the level A packaging shall be examined in accordance with the classification of defects and with MIL-STD-105. During each hour of production, a sample of ignition mix shall be drawn from the outlet of the ignition mix hopper and tested in accordance with 4.2.4.3.

4.2.3.2 For destructive test. Sample fuzes shall be tested in accordance with 4.2.4. Disposition of the lot represented by the sample fuzes shall be determined in accordance with MIL-STD-105, using an AQL of 1.0 percent defective. Disposition of the lot represented by the sample containers shall be determined by using an AQL of 1.0 percent defective in lieu of the 4.0 percent defective specified in MIL-P-116.

4.2.3.3 Critical characteristics. Each fuze in the lot shall be examined for its critical characteristic listed in the classification of defects.

4.2.3.4 Classification of defects.

(a) Fuze, smoke pot, floating, electric, M209 (Drawing 36-7-25).

Categories	Defects	Acceptance standard
Critical:		
1	Wire not shunted	
Major:	AQL 1.0 percent defective	
101	Component missing or incorrectly assembled	
102	Thread incorrect	
103	Torque, cap to housing incorrect	
104	Marking incorrect	
105	Parrafin dip and shellac coating missing (discharge end)	
106	Electrical resistance incorrect	4.2.4.2
107	Workmanship (see 3.6)	
Minor:	AQL 2.5 percent defective	
201	Packaging component missing or incorrectly assembled	
202	Contents loose in container	
203	Marking missing or illegible	

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(b) Packaging, fuze (Drawing 36-3-114).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Fuze missing
102	Marking incorrect
Minor:	AQL 2.5 percent defective
201	Packaging component missing or incorrectly assembled
202	Contents loose in container
203	Marking missing or illegible

4.2.4 Tests.

4.2.4.1 Functioning. The fuze shall be submerged to a minimum depth of 6 inches in water maintained at a temperature of 77° ± 9°F for a period of 4 + 1/2 hour. The fuze shall be mounted in a fixture (figure 1) and functioned, utilizing a do current of 1.0 amp maximum with a potential of 1.5 volts maximum when applied for 100 milliseconds maximum in such a manner that the flash is directed downward.

4.2.4.2 Electrical resistance The soldered shunt shall be removed. The fuze shall be tested with the Alinco Tester No. 105-5A or other type of tester approved by the Government (see 6.5).

4.2.4.3 Dryness. The moisture and volatile content of the sample shall be determined in accordance with MIL-STD-1234, method 101 or 102.1.

5. PACKAGING

5.1 Unit packing. Level shall be as specified (see 6.2).

5.1.1 Level A. Ten fuzes shall be unit packed together in accordance with Drawing 36-3-114.

5.1.2 Level B. Unit pack shall be as specified in 5.1.1.

5.2 Packing. Level shall be as specified (see 6.2).

5.2.1 Level A. Thirty cans of fuzes, unit packed as specified in 5.1.1, shall be packed in a wooden shipping container conforming to either type overseas, style J of PPP-B-601, class 2, style 4 of PPP-B-621, or type I, class 2, grade A of MIL-B-46506. The shipping containers and all wood parts thereof shall be completely immersed for a minimum of three minutes in a water repellant

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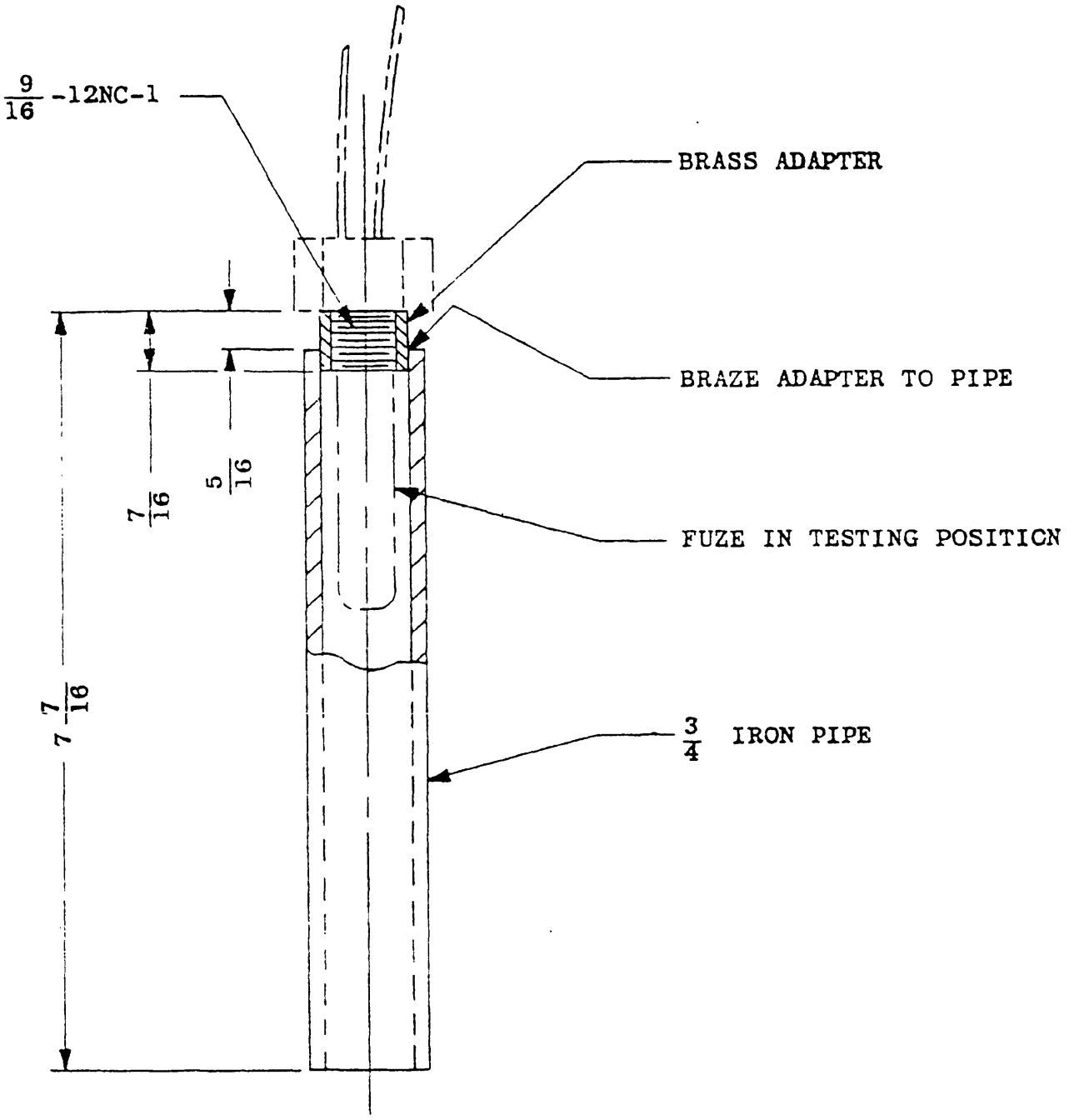


FIGURE 1. Fuze testing fixture.

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wood preservative conforming to composition B of TT-W-572. Prior to packing the cans into the treated shipping container, the container shall meet the dryness-of-preservative requirement specified in MIL-B-2427. The inside dimensions of the container shall be approximately 15 by 13-1/4 by 8-3/8 inches (length, width and depth respectively). The 30 cans of fuzes shall be packed in the container in 2 tiers of 15 per tier. Fiberboard pads fabricated from material conforming to type CF, class weather resistant, variety SW of PPP-F-320 shall be used between the contents and the sides, ends, top and bottom of the box; and between the 2 tiers of canned fuzes. Additional pads shall be used as required to make a tight pack. Also, each tier in the box shall De fitted with interlocking slotted fiberboard partitions made from the same material as the fiberboard pads. Closure of the container shall be in accordance with the requirements of the applicable container specification and its appendix using galvanized strapping.

5.2.2 Level B. Fuzes shall be packed in accordance with 5.2.1 except the wood preservative is not required.

5.2.3 Interplant shipment (see 6.4). The quantity of fuzes specified in the contract or order (see 6.2) shall be prepared for shipment in containers which comply with the Code of Federal Regulations and the rules and regulations applicable to the mode of transportation. Shipping containers shall assure safe delivery to destination without damage to the contents. Interplant shipment pack shall not be shipped on Government Bill of Loading (GBL).

5.2.4 Unitized loads. When specified in the contract or order, level A and B snipping containers shall be palletized in accordance with Drawing 19-48-4116,/ 100AR1.

5.3 Marking.

5.3.1 Levels A and B. Marking shall be in accordance with MIL-STD-129. In addition to any specific marking required by the contract or order, unit packs shall be marked in accordance with Drawing 36-3-114. Shipping containers shall be marked in accordance with Drawing 36-7-44.

5.3.2 Interplant shipment (see 6.4). Containers shall be marked in accordance with FED-STD-123 and the Code of Federal Regulations.

6. NOTES

6.1 Intended use. This item is used in burning-type munitions.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Level of unit packing and packing required (see 5.1 and 5.2).
(c) Quantity required for interplant shipment (see 5.2.3).

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6.3 Caution. Precaution should be taken to prevent fires or explosions incident to the preparation of the ignition mixture while assembling the fuzes. Adequate ventilation should be provided in the area where the pyrotechnic mixture is prepared and also in the area where the destructive tests are conducted.

6.4 Interplant shipment. Packaging for material which does not enter the military supply system such as shipments from vendor to a contractor, a subcontractor, or a supplier to a military arsenal or plant.

6.5 Resistance tester. This instrument may be obtained from the Alleghany Instrument Company, Cumberland, Maryland.

6.6 Subject term (key word) listing.

Fuze
Ignition mix
Smoke pot
Squib

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 1365-A092

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-11673F(EA)	FUZE, SMOKE POT, FLOATING, ELECTRIC, M209

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NOTICE OF INACTIVATION FOR NEW DESIGN

INCH-POUND

MIL-F-11673F(EA)
NOTICE 1
19 April 1996

MILITARY SPECIFICATION

FUZE, SMOKE POT, FLOATING, ELECTRIC, M209

This notice should be filed in front of MIL-F-11673 F(EA) dated 17 December 1987.

MIL-F-11673F(EA) is inactive for new design and is no longer used except for replacement purposes.

Custodian:

Army - EA

Preparing activity:

Army - EA

(Project 1365 -A122)

Review Activities:

Airi Force - 99

AMSC N/A

FSC 1365

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